Three Additional Shutter-SOURCE Calculations v0.2

Supply averaging, photon pulses, and CMB non-Gaussian correlations

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VALIDATION INPUTS

c=299792458 m/s, dt=1.4813019664158633e-21 s, ell=0.4440831575520451 pm. Use all 1,015,000 addresses with the inherited address-1 exclusion, residue and supplier ledgers, 128 finite preparation states, and the lambda_D=0.25 candidate. 676,666 addresses have controlled supplier bits. Mean dark-profile RMS is 1.2107407781339933e-8; return-profile RMS is 9.489398775439113e-9. Use actual cached patched-CAMB D/R transfer arrays for T,E,phi: seven k=0.05*n Mpc^-1, n=1..7, every ell=2..6000, accuracy=2.5. Freeze input hashes. No new observational fit or Boltzmann run is performed. Higher moments are actually calculated through the stored physical transfer operators.

WRRA TRANSFORMATION

A. Intrinsic supplier noise, temporal covariance, and resources

Recompute address-weighted conditional Bernoulli variance using actual p_r(J). One-frame relative RMS: D=0.3451400342, R=0.4941525318. Actual ownership gives zero intrinsic D-R supplier covariance; shared phase mean-stress covariance is retained separately. Compare explicitly declared temporal laws: r=0: unchanged bits; r=1: independent preparations each physical shutter; 0<r<1: stationary renewal P=(1-r)I+r*1*pi with correlation (1-r)^lag. Var(average)/Var(one)=[N+2 sum_h(N-h)(1-r)^h]/N^2. Static bits do not average down. The initial phase register remains static and its mean-profile variance remains as a floor, distinct from intrinsic noise. Count fresh bit preparations, initial/state entropy, path-record entropy and conditional absolute energy turnover. Opposite reservoir accounting is required; absolute turnover is not equated with irreversible depletion or heat dissipation. An independent 60,000-bit, 128-step Monte Carlo verifies the correlation variance and unit-gap turnover. No claim of directly sampling 10^15 physical shutters.

B. Photon propagation and pulses

Use the inherited exact shift plus fixed optical model 1.0: static phase label, g=1, phi(E,J)=E/hbar*(L/c)*f_D(J). Its group delay is (L/c)*f_D(J) and its energy phase curvature is zero. No additional energy-dependent dispersion is inserted. For a declared 1% intensity-energy bandwidth, initial Gaussian sigma_t=hbar/(2 sigma_E). Nonselective pulses are exact mixtures of 128 shifted Gaussians with variance sigma_t^2+Var(delay). Verify the 1 keV, 1 m positive-frequency FFT pulse, normalization, energy conservation and timing variance. Other bands use exact finite-mixture moments and CDF bins. Audit high-energy phase aliasing with matched time/space refinement; do not infer a physical gamma-ray cutoff.

C. Higher SOURCE moments through actual CMB transfers

Adopt one shared spatial phase for seven collinear harmonics and a uniformly rotated common axis. This joint preparation is explicit; 2-point spectra alone do not select it. Represent orientation by a finite weighted quadrature with 12,001 mu nodes and 24,001 azimuth nodes: 288,036,001 axis states. Joint with the 128 phases: 36,868,608,128 states, entropy 24.18593808 nats. This exactly represents orientation moments through fourth order for ell<=6000. Count the additional ensemble representation and initial preparation; do not assume free repeated or infinite orientation memory. Natural preparation dynamics remain separate. Apply actual T,E,phi transfers to cos/sin coefficients, generating axial a_l0(J). First reproduce C_l=<a_l0^2>/(2l+1), TT/EE/TE and phi normalization. Rotational B_l1l2l3=Wigner(000)*<a_l10 a_l20 a_l30>, with reduced b=B/Gaunt factor. Calculate selected allowed TTT, EEE, TTE, TEE, TTphi, TEphi, phiphiphi triangles and every equilateral ell=2..6000. This is not all triangles through ell=256. Calculate connected angular power covariance as exact finite-phase covariance minus its Gaussian Wick term, and sky-point cumulants for bands ending at ell=32,64,128,256. These are actual 4-point statistics, not an exhaustive trispectrum. Check 64/128/256 phase quadrature, finite-angle normalization, and independent sky cubic moments versus bispectrum contraction. Distinguish an isotropic Gaussian ANGULAR field from Gaussian-moment INITIAL harmonics sharing one random axis. The latter has a mixture of orientation-dependent conditional covariances and need not have zero connected fourth moment. Two-point power alone does not select the joint preparation or determine its connected higher moments.

OUTPUTS

A. Intrinsic-noise reduction to the inherited mean-profile RMS

Policy Dark D Return R
Static r=0 No crossing No crossing
Independent r=1 1.203737902 microseconds 4.016874415 microseconds
Renewal r=1/128 306.953164899 microseconds 1.024302976 milliseconds
Independent thresholds require approximately 5.4987e20 / 1.8349e21 fresh bit preparations. At r=1/128 this increases to 1.0955e21 / 3.6555e21.
Absolute energy-turnover density rates: 3.332895850e11 W/m^3 for r=1 and 2.603824883e9 W/m^3 for r=1/128. These are conditional ledger turnover, not measured heat dissipation or power consumption. No reservoir budget is supplied, so the averaging mechanism is not certified as resource-feasible. dt remains in T=N*dt.
One-frame algebraic local linear-Friedmann sensitivity is 0.17512207 relative RMS; do not treat this large-fluctuation diagnostic as accurate stochastic GR dynamics.
A conserved reservoir stress tensor, spatial noise construction and dynamical gravitational noise spectrum are not executed.

B. Photon outputs

Extra energy-dependent mean delay=0 and intrinsic dispersion=0 for the exact free shift and fixed linear SOURCE phase rule. At 1 m differential exposure, SOURCE arrival-time RMS=4.038596522e-17 s=40.38596522 attoseconds. Each fixed label merely shifts its pulse; nonselective mixing broadens it. Equal common-mode coupling cancels relative effects. At fixed fractional bandwidth, higher-energy prepared pulses are shorter, so relative broadening changes; this is not vacuum dispersion. Base phase-alias energy is 1.3959569995 MeV with wavelength 0.8881663151 pm: representational audit only, not a physical cutoff.

C. CMB outputs

Initial dark profile: skewness 0.77344541, excess kurtosis -0.93821798. Actual transferred SOURCE temperature statistics for ell=2..256: RMS 0.0005034727028 microK; skewness -1.1903507147; excess kurtosis 32.2551732492; third moment -1.5191572272e-10 microK^3; connected fourth 2.0725408293e-12 microK^4. Initial-profile and projected-sky skewness are distinct, including a sign change. With an independent unlensed Gaussian reference of RMS 93.23950613 microK: combined skewness -1.8741434548e-16; excess kurtosis 2.7422259540e-20. This reference is not a full lensed/foreground NG likelihood. Largest equilateral |reduced TTT b| is 3.1800803116e-13 microK^3 at ell=2. SOURCE deflection RMS 0.009556588544 arcsec is reproduced. No observational detection of the extra component is claimed.

REFUTATION CONDITIONS

Revise incorrect temporal independence, averaging of unchanged bits, unaccounted fresh preparation/recording/reservoir flux, failed total-ledger conservation, incompatible fixed phase/pulse law, or phase-alias artifacts claimed as physical signals. Revise failed CAMB/Gaunt normalization or finite quadrature. Compare the fixed custom CMB shape to observations before an observational claim. Known-data calibration is legitimate; identify changed rules as a new version.

ASSESSMENT

All three targets are calculated, with 176 numerical controls passed. Supply averaging is conditional on correlation and resources. Existing photon rules give zero intrinsic energy dispersion and a nonselective arrival distribution. Higher SOURCE statistics survive actual transfer but are strongly diluted by the reference CMB. Observational confirmation, full stochastic GR for supplier noise, a custom NG likelihood and exact non-Gaussian lens remapping remain distinct tasks.

REPRODUCTION

python -m pip install -r requirements.txt
OPENBLAS_NUM_THREADS=1 python reproduce.py
python figures.py
python verify.py
All input hashes and assumptions reside in inputs/frozen_tests.json. Results
contain JSON, CSV, NPZ and a summary figure. Release DOI: 10.5281/zenodo.23380920. The prior clock-to-CMB record is retained.
python verify.py --archive checks all distributed bytes before rerunning.
Default verify checks frozen inputs/source and stable numerical observables after
a rerun; output wall-clock timing is excluded from numerical regression.

V0.2 REVIEW AND CORRECTIONS

The main v0.1 scientific observables are unchanged. The preserved v0.1 reports under review/ are historical evidence, superseded by this report. 1. A finite Gaussian-moment harmonic comparator uses 14 independent ternary variables with nodes 0,+sqrt(3),-sqrt(3), weights 2/3,1/6,1/6. It matches Gaussian moments through fourth order and reproduces two-point power. Its 3^14=4,782,969 states do not constitute a full Gaussian distribution or a unique physical law. At ell=2..256, common-axis comparator excess kurtosis is 40.7442439509 (T), 48.5937322940 (E), 8.2394888353 (phi); the original SOURCE values are 32.2551732492, 22.8054297802, 9.7416539350. An isotropic Gaussian angular field instead has zero connected higher cumulants. Common-axis mixing itself therefore contributes to non-Gaussianity; the signal is not unique evidence of quantization. 2. Group-delay dispersion is hbar_eV^2*d^2(phi)/d(E_eV)^2 in s^2. It remains zero. The transfer is a fixed-duration phase screen at nominal L/c; no independent bulk-medium traversal-time law is derived. 3. Recorded-history entropy is conditional on actually keeping that history. It is not a universal thermodynamic erasure cost or a mandatory memory bound. The large local linear-gravity RMS is outside a controlled small-noise regime. 4. Temporal-domain checks and the already-below-target threshold edge case are fixed. Independent 75-digit Decimal checks cover 42 correlation/count cases. 5. Byte-integrity verification and stable numerical regression are separated. All 176 controls pass: supply 39, photon 16, CMB 65, internal review 56. Passing controls establishes these stated numerical properties, not empirical confirmation, unique physical derivation, or an independent external assessment.

PRIMARY COMPARISON SOURCES

CAMB transfer conventions: https://camb.readthedocs.io/en/stable/results.html Planck NG conventions: https://arxiv.org/abs/1905.05697 Stress-correlation/gravity-noise methodology: https://arxiv.org/abs/0802.0658 Detailed original input provenance and optical rules are retained in inputs/.

Numerical overview







